DESIGN COMPUTED · FREE GUIDE

How a burned monument is rebuilt

The free guide to the reel on the Limoges town hall: what restorers rebuild from, how Notre-Dame came back, who records heritage, and a first exercise to understand how a building is surveyed.

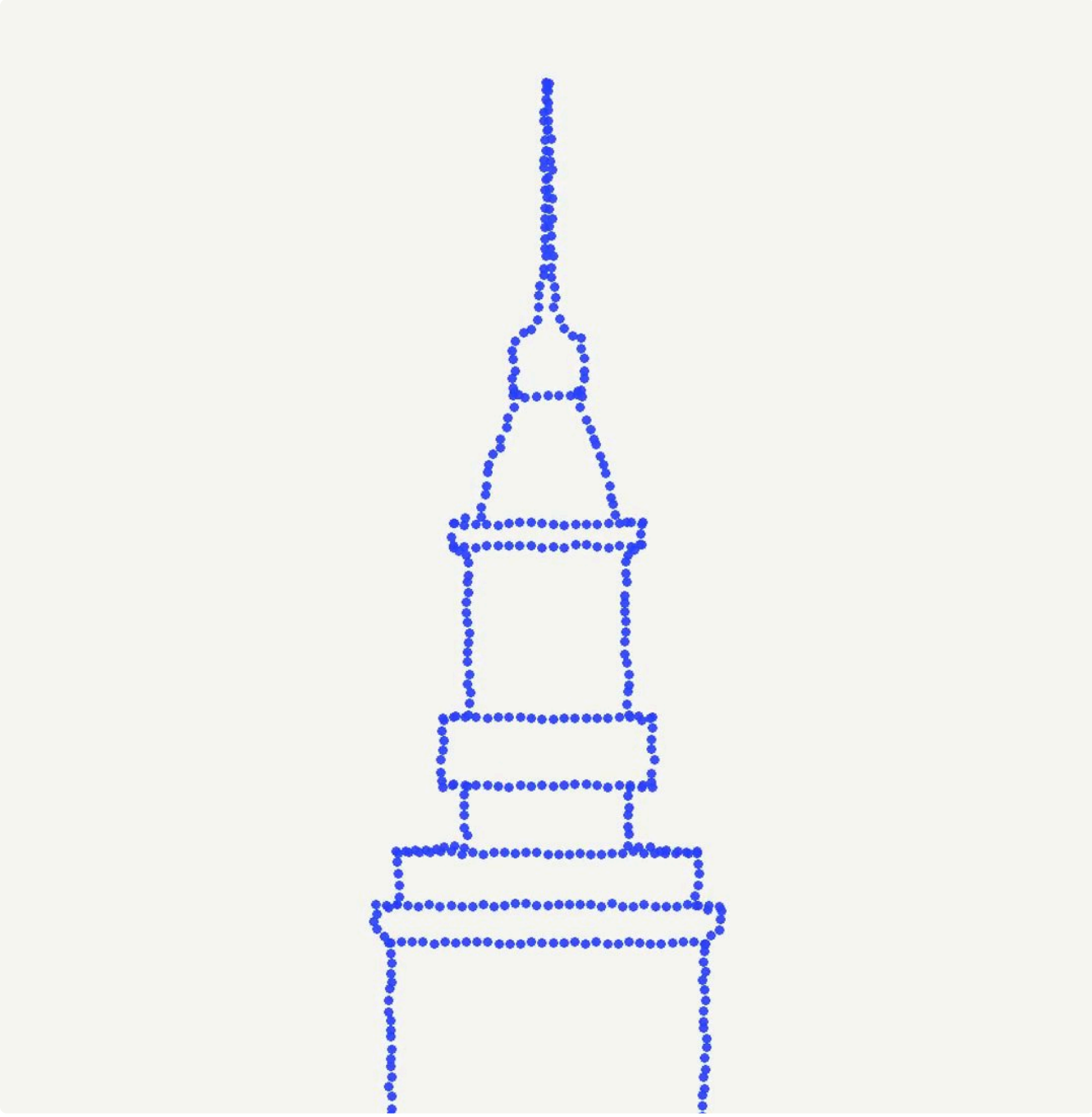


What restorers rebuild from

A monument is rebuilt from its records. Archives hold the original drawings, specifications and repair files. Old photographs and postcards show details the drawings leave out. Measured surveys give exact dimensions and shapes.

Today, professional surveyors record buildings with laser scanners and photogrammetry: millions of measured points that describe every surface. For listed monuments in France, this work is led by specialised heritage architects and the State's heritage services.

No single source is enough. Restorers cross-check drawings, photos, surveys and the remains of the building itself before anything is rebuilt.



The Limoges bell tower drawn as a point cloud: an illustration of what a survey records.

Why it saves monuments

At Notre-Dame de Paris, restorers could rely on archives, the historic drawings, and the laser survey made years earlier by the art historian Andrew Tallon. The cathedral reopened in December 2024, five years after the 2019 fire.

The Hôtel de ville of Limoges (Charles-Alfred Leclerc, inaugurated on 14 July 1883, paid for by the bequest of Alfred Fournier, listed in 1975) was set on fire on 17 October 2026, at the end of a protest. Its 43-metre bell tower collapsed. Its future is now in the hands of the city, the State and the heritage experts.

Who scans heritage

Notre-Dame de Paris (2010s)	Andrew Tallon's laser scans of the cathedral, used after the 2019 fire.
CyArk (since 2003)	A non-profit that records endangered heritage sites in 3D and shares the data.
Zamani Project	University of Cape Town: 3D documentation of heritage sites in Africa and beyond.
Rekrei / Project Mosul (2015)	Volunteers rebuilt lost objects in 3D from tourists' photos.
Historic England	Publishes free guidance on photogrammetry for heritage recording.

A first exercise: survey a small detail (free)

1. This exercise is to understand how surveying works. It does not replace a professional survey of a monument.
2. Pick a small subject: a doorway, a carved stone, a fountain.
3. Choose soft light: an overcast day avoids hard shadows and burnt highlights.
4. Install a free app (RealityScan or the free tier of Polycam on a phone), or take photos for Meshroom on a computer.
5. Walk slowly around the subject and take a photo every step, with each photo overlapping the previous one by about two thirds.
6. Do a second ring from higher or lower, and close-ups of the details (carvings, joints).
7. Avoid glass and very shiny surfaces: they confuse the matching. Keep the subject still and sharp.
8. Let the software build the model, then scale it with one measured distance.
9. Export it (OBJ or GLB) and open it in Rhino or Blender: now you can draw sections, elevations and details from it.

The rule to remember

A building can only be rebuilt as well as it was recorded. Archives, photos and surveys made today are what restorers will work from tomorrow.

Mistake 1: too few photos	Gaps in overlap leave holes. More photos is almost always better.
Mistake 2: no scale	Without one known distance the model is a nice picture, not a measurement.
Mistake 3: one height only	Shoot at least two rings at different heights, or the top and underside are missing.

Go further

Photogrammetric Applications for Cultural Heritage — Historic England (2017)
https://historicengland.org.uk/images-books/publications/photogrammetric-applications-for-cultural-heritage/

CyArk — 3D records of heritage sites
https://www.cyark.org

Meshroom (free, open source) — AliceVision
https://alicevision.org

Hôtel de ville de Limoges — Wikipedia (FR)
https://fr.wikipedia.org/wiki/H%C3%B4tel_de_ville_de_Limoges

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